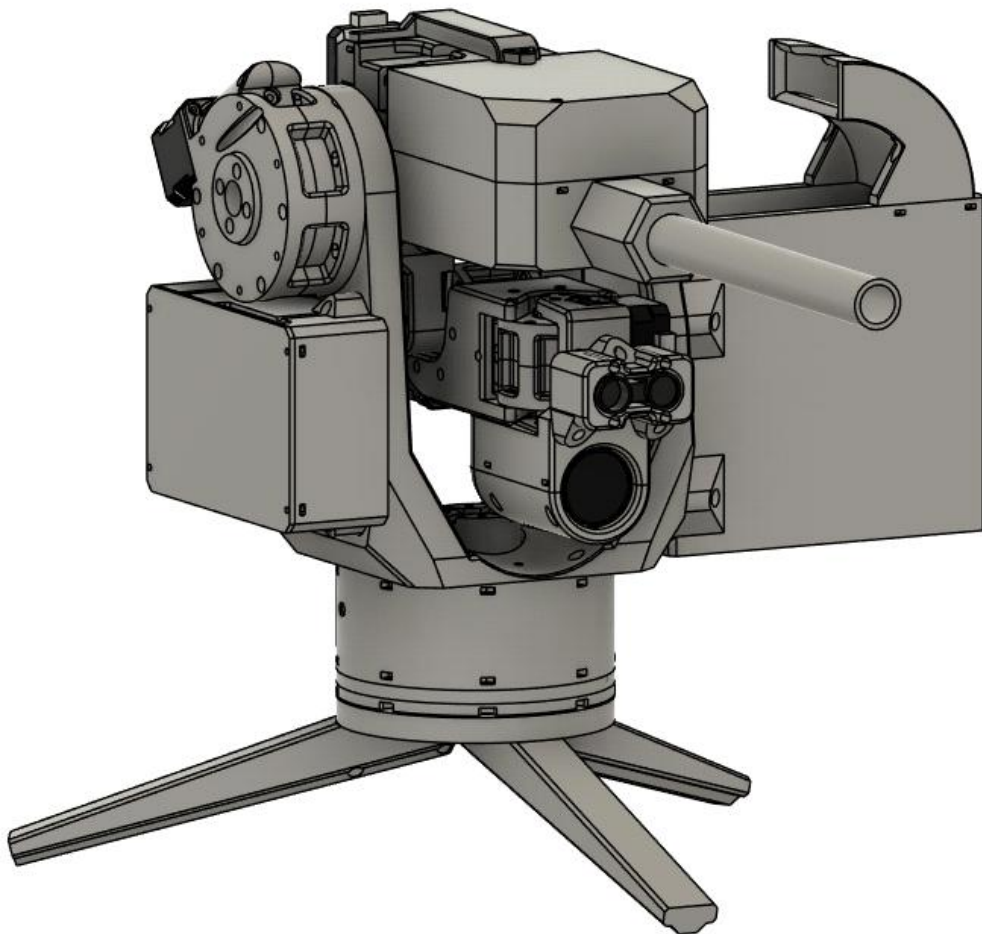


Mechanical Design Handbook

3D Printed Remote Controlled Weapon System v5



WILDCARD

Table of contents

1. Overview

2. System Architecture

3. Drive system

3-1. Main frame & Neck frame

3-2. Main actuator

3-3. 3D printed azimuth bearing

3-4. Tripod

4. Launcher

4-1. Feeder

4-2. Accelerator

5. EOTS

5-1. EOTS actuator

5-2. Camera frame

6. Electronic control

1. Overview

RCWS(Remote controlled Waepon System) v5 prototype is one of several 3D-printed RCWS projects. The goal of This projects is to demonstrate an electro-optical fire control system at a range of approximately 30 meters. The RCWS v5 prototype was designed so that all mechanical parts, except for electronic components and bearings, can be fabricated using an Ender 3 v2 3D printer.

This document presents the mechanical design and assembly of the RCWS v5.

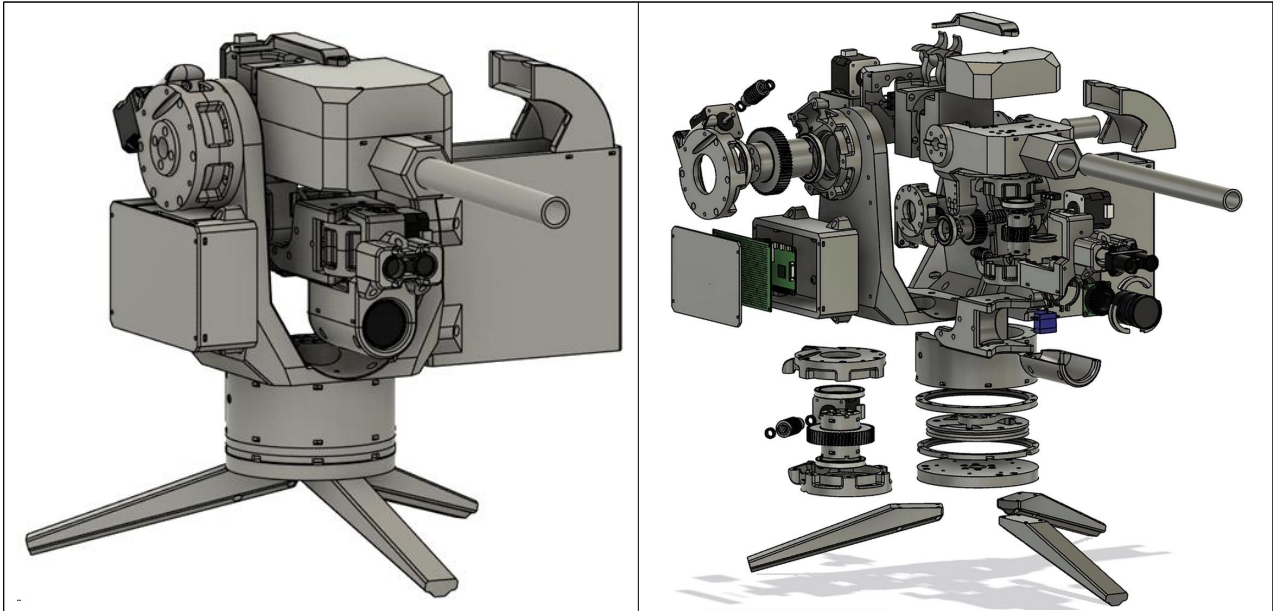


Figure 1.

2. System architecture

Structure of RCWS v5 can be categorized into four main sub system: drive system, launcher, EOTS, and electronic control.

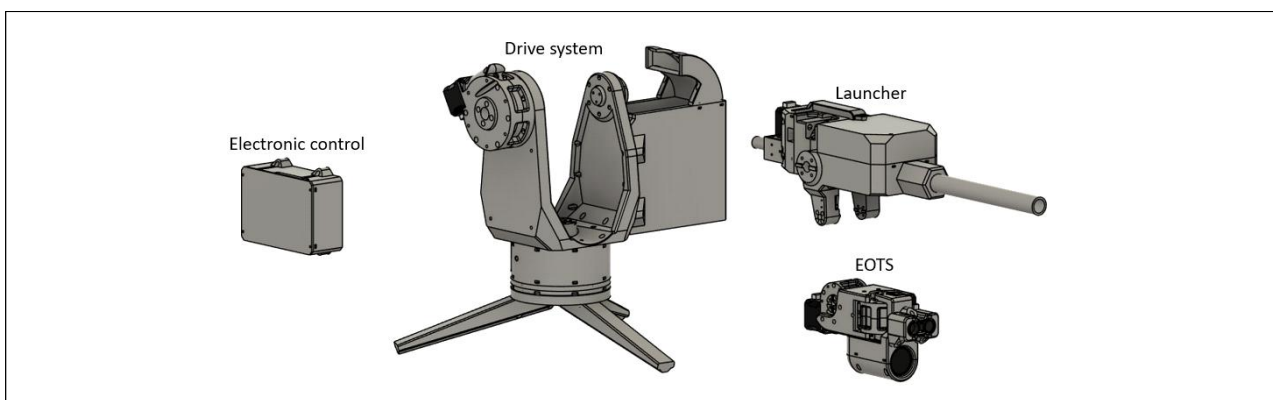


Figure 2.

The drive system and EOTS has each use two worm-gear actuator. The launcher and camera can be controlled independently. This four-axis architecture satisfies the fire control requirements for dynamically moving target and long-range firing.

All part of the RCWS v5 can be printed using an Ender3 v2 3D printer. The largest part is Main_Frame, with a bounding box size of 180 x 145 x 248mm.

3. Drive system

The drive system is a two-axis pan-tilt chassis comprising the main frame, neck frame, two worm-gear actuators, a 3D printed azimuth bearing, and a tripod.

3-1. Main frame & Neck frame

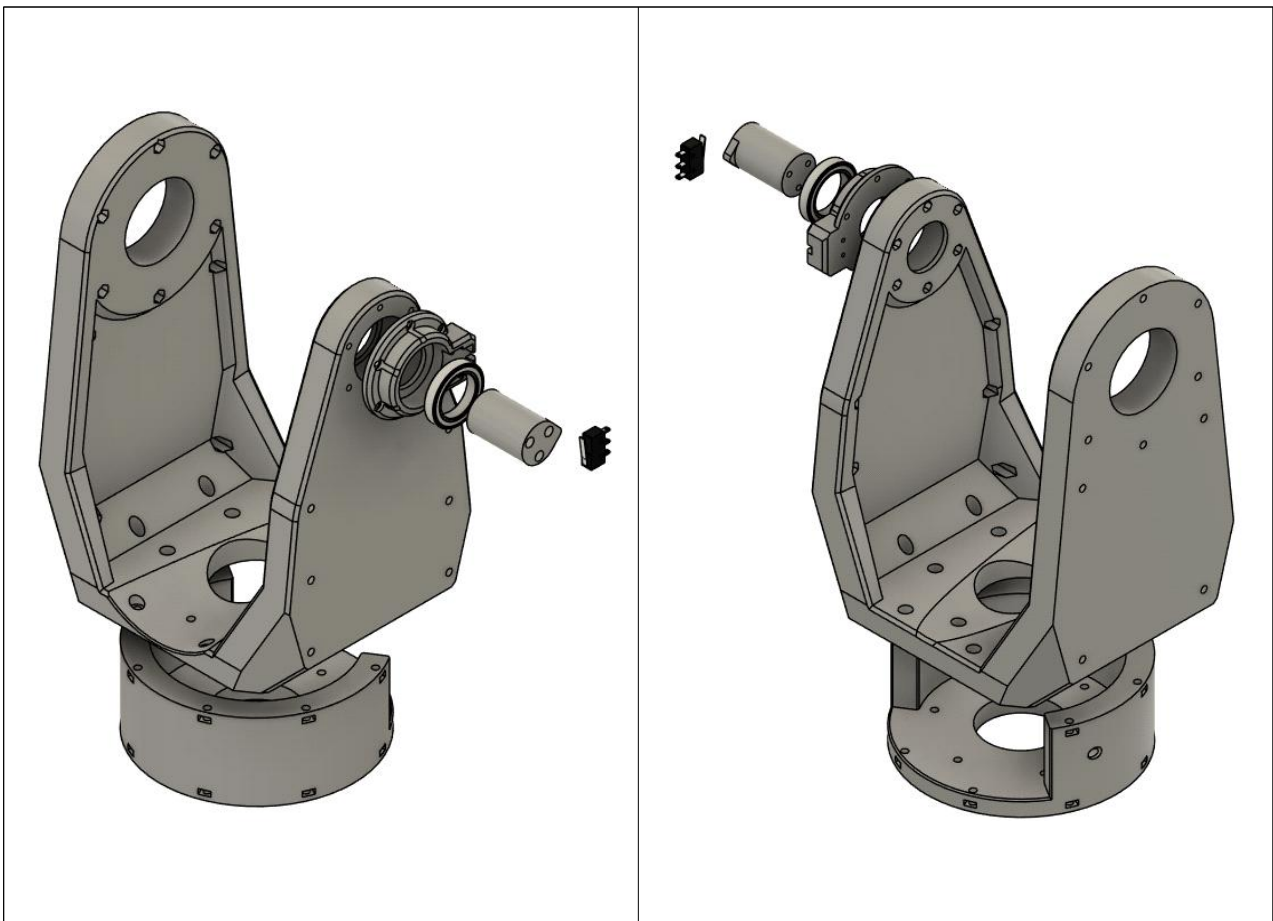


Figure 3.

<Non 3D printed Parts>

M2 x 10mm bolt

M3 x 10mm bolt

M3 x 12mm bolt

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M3 hex nut
M4 x 15mm bolt
limit switch
6805ZZ bearing

3-2. Main actuator



Figure 4.

<Non 3D printed Parts>
NEMA17 stepper motor

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M3 x 5mm bolt
M3 hex nut
M4 x 15mm bolt
M4 hex nut
6809ZZ bearing
MR128ZZ bearing

3-3. 3D printed azimuth bearing

The entire frame and beads are 3D printed because of cost of bearing.

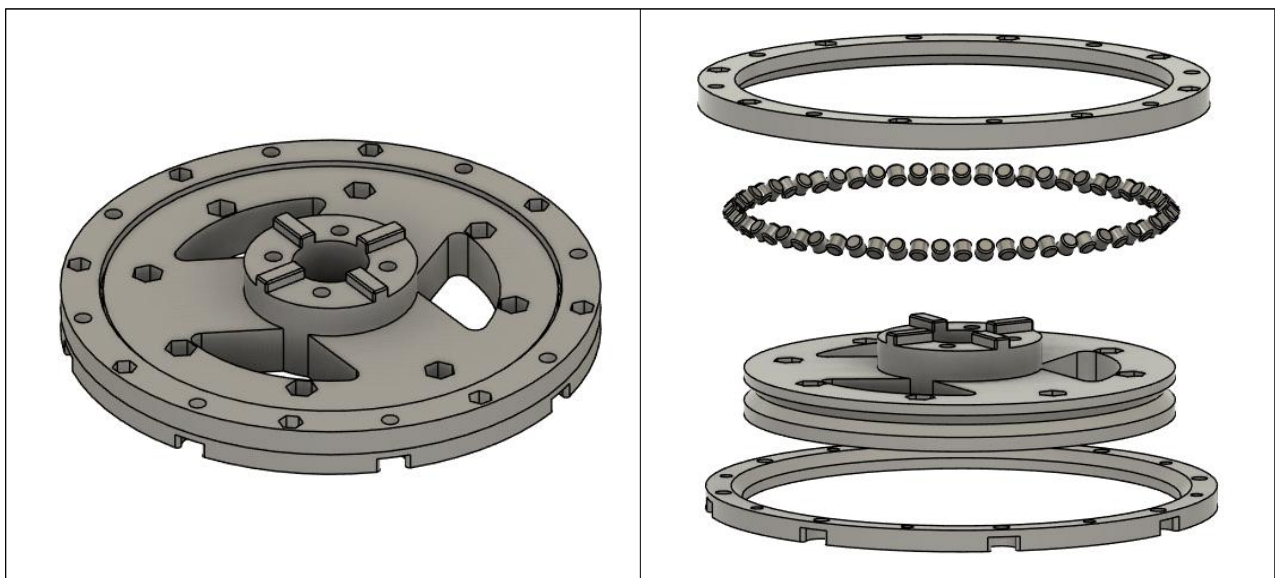


Figure 5.

<Non 3D printed parts>

M3 x 12mm bolt
M3 hex nut
M4 x 15mm bolt
M4 x 20mm bolt
M4 hex nut

3-4. Tripod

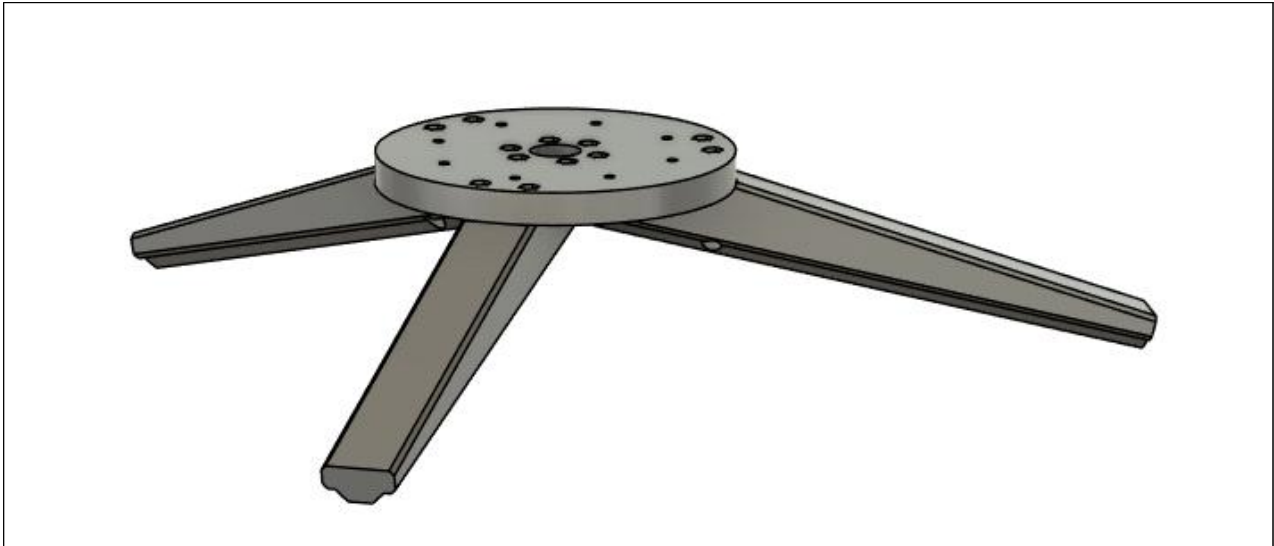


Figure 6.

<Non 3D printed parts>

M4 x 15mm bolt

M4 hex nut

4. Launcher

The launcher primarily consists of a feeder and an accelerator. An EOTS mount is located on the lower section of the assembly.

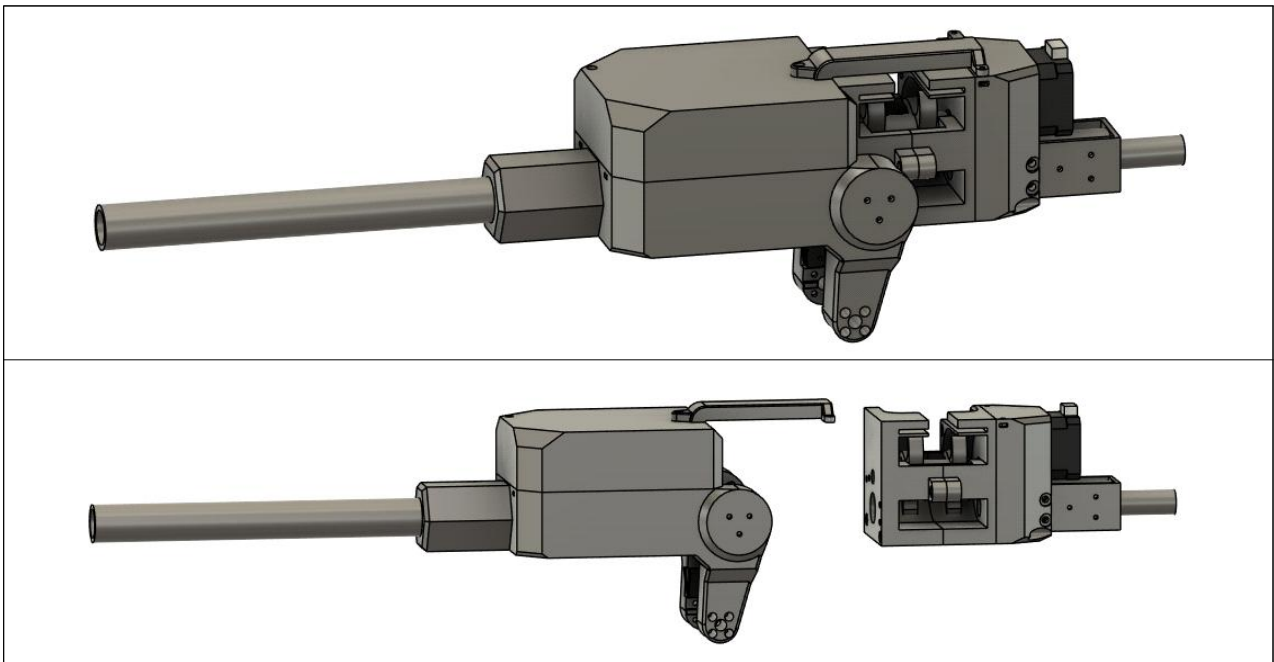


Figure 7.

4-1. Feeder

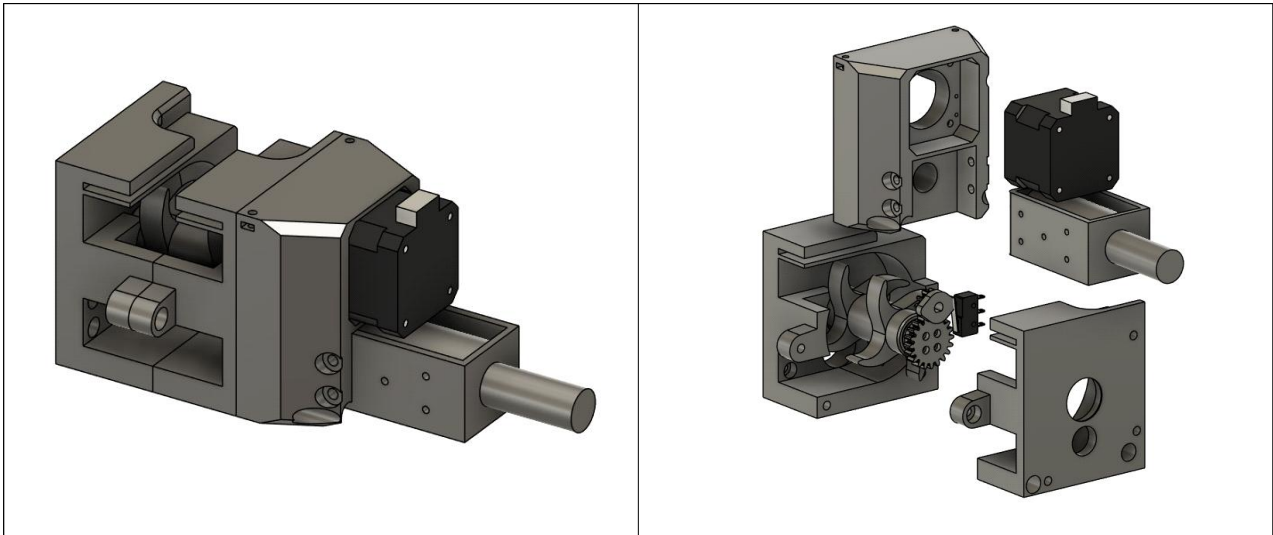


Figure 8.

<Non 3D printed parts>

NEMA17 stepper motor

35mm stroke 12v solenoid

M2 x 10mm bolt

M3 x 5mm bolt

M3 x 10mm bolt

M4 x 15mm bolt

M4 hex nuts

4-2. Accelerator

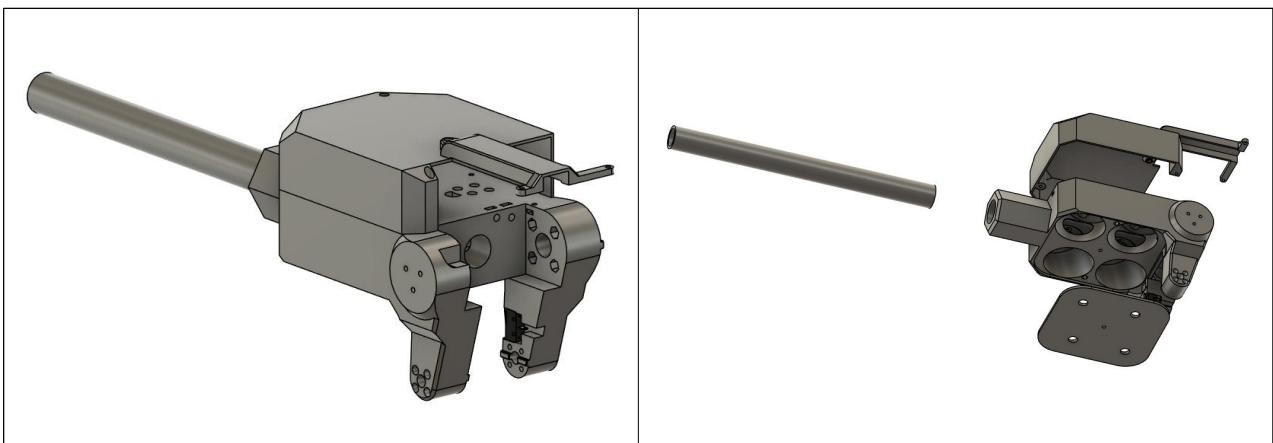


Figure 9.

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<Non 3D printed parts>

2204 2300kv BLDC motor

OD 26mm ID 20mm pvc pipe

limit switch

M3 x 7mm bolt

M3 hex nut

M4 x 15mm bolt

M4 hex nut

5. EOTS

The EOTS features a cable-driven camera zoom mechanism and an adjustable rangefinder zero. It is mounted on a two-axis gimbal, and each axis can be calibrated using an individual limit switch.

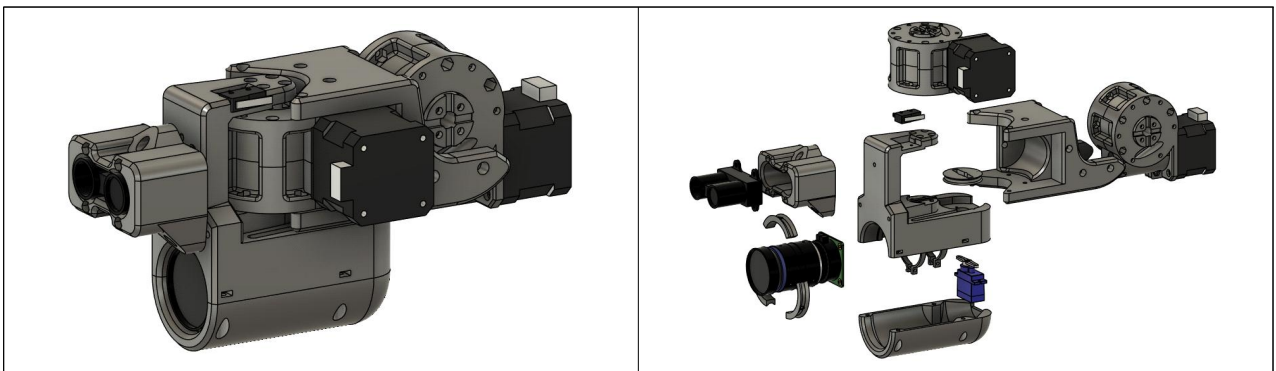


Figure 10.

5-1. EOTS actuator

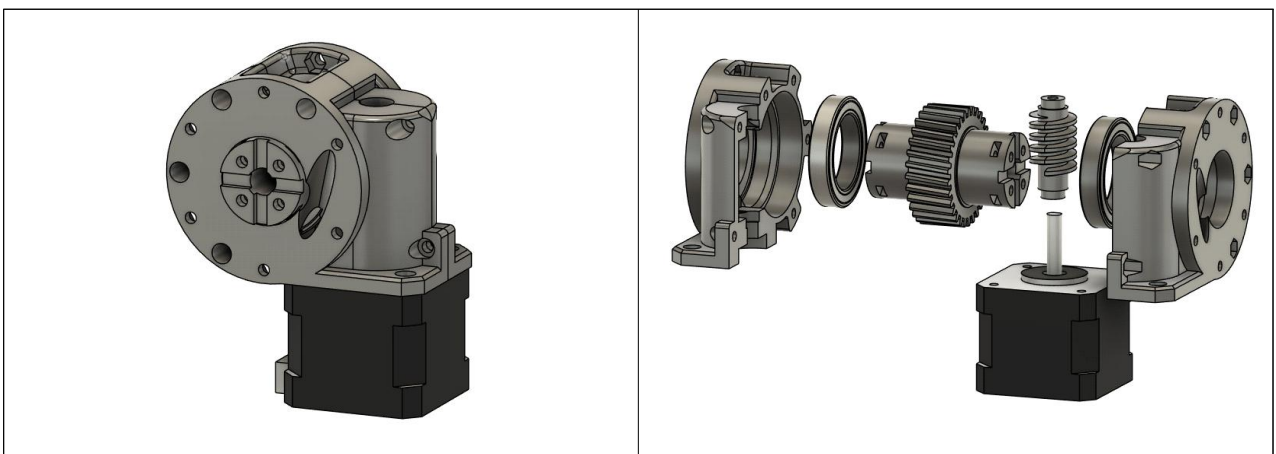


Figure 11.

<Non 3D printed parts>

M3 x 5mm bolt

M3 x 10mm bolt

M3 hex nut

MR128ZZ bearing

6805ZZ bearing

5-2. Camera frame

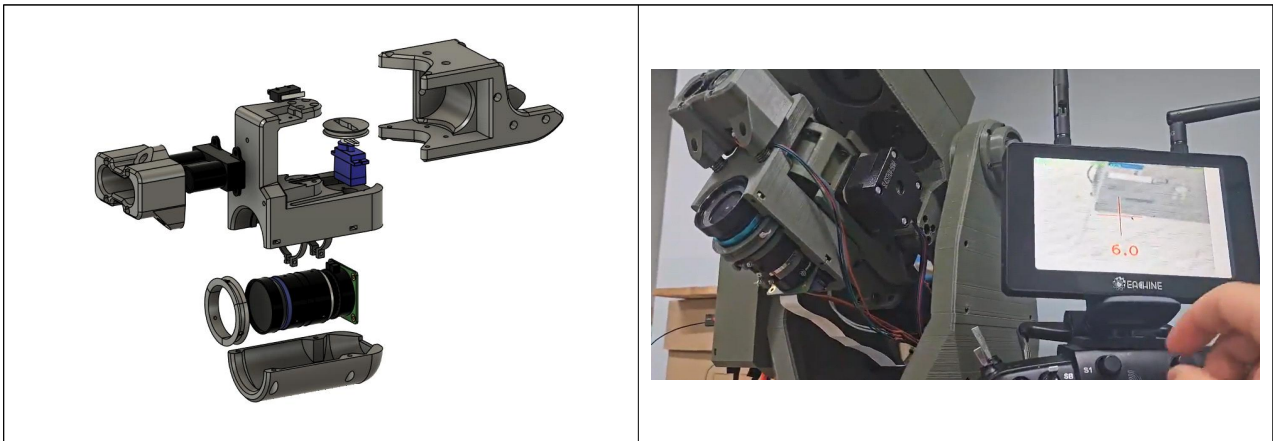


Figure 12.



Figure 13.

<Non 3D printed parts>

sg90 servo motor

raspberry pi HQ camera

limit switch

M2 x 3mm bolt

M2 x 12mm bolt

M2 hex nut

M3 x 10mm bolt

M3 x 25mm bolt

M3 hex nut

6. Electronic control

Unfortunately, most of the documentation related to the electronic control system has been lost. An Arduino was used to control the stepper motors and interpret PWM signals from the RC transmitter and receiver, while a Raspberry Pi 4 was used for image processing.

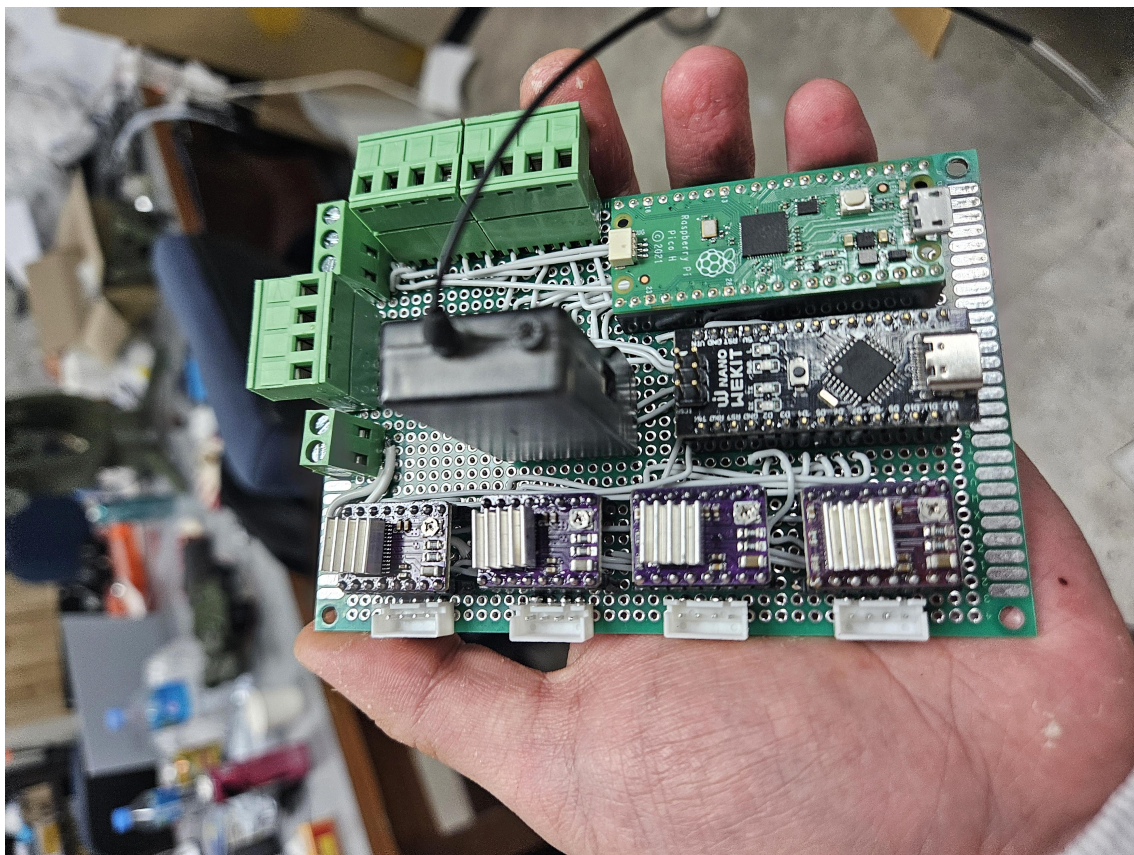


Figure 14.

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Figure 15.